Auto-Decte: Trust-Constrained Document Intelligence through Candidate–Fact Separation and Certificate-Bound Human Authorization

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Abstract

Document intelligence in payroll-adjacent and production workflows must control both recognition error and the authority granted to machine output. Auto-Decte is a trust-constrained human-in-the-loop architecture for structured industrial forms. Its central invariant separates immutable evidence, append-only machine candidates, attributable human decisions, and versioned facts: perception, retrieval, and language-model services have no direct fact-writing capability through the evaluated application interfaces. The implemented authority contract makes this invariant executable: every recognition result is persisted as a content-addressed candidate certificate bound to its field, evidence, and expected fact version; a confirmation is a per-field attributable human decision over a database-loaded certificate, committed with the record version, decisions, transitions, and audit event in a single compare-and-swap-protected transaction; and reverse traces rebuild each field chain from evidence through certificate, decision, and transition, reporting any tampered, missing, or wrongly bound entry as incomplete.

We evaluated the implementation at three levels. First, a seed-disjoint cell benchmark measured recognition: a HOG–linear-SVM recognizer achieved 99.81% accuracy at full coverage, and 60 synthetic forms under five whole-page conditions achieved 60% complete-form success, exposing QR and marker failures under blur and rotation. Second, an executable authority-integration benchmark ran the real services over a synthetic fault corpus with a scripted reviewer: every injected corruption (tampered transition producer, deleted transition, rebound record-version row) was detected as an incomplete trace, cross-field certificate substitution and cross-form evidence were rejected before any write, all valid certificate confirmations succeeded, and stale replay was contained. Third, the integrated software verification suite passes 239 automated tests, and the benchmark outputs are byte-identical across repeated runs under an immutable manifest. The evidence supports the executable authority boundary but not production effectiveness: independently transcribed real forms and controlled reviewer studies remain necessary.

Keywords: document intelligence, human-in-the-loop, selective prediction, data provenance, trustworthy artificial intelligence

1. Introduction

Document processing is often evaluated as a prediction problem: an image enters a model and structured values leave it. That abstraction is incomplete in payroll-adjacent and production workflows. A wrong digit may alter a wage calculation, a stale record may be mistaken for current evidence, and a plausible language-model suggestion may be granted authority that its provenance does not justify. In such settings, the consequential question is not only whether a machine output is accurate, but also whether that output can silently become an operational fact.

Modern document models substantially improve representation learning and extraction across layouts (Xu et al., 2020, 2021; Huang et al., 2022; Kim et al., 2022; Appalaraju et al., 2021; Li et al., 2023). Human-in-the-loop systems can direct difficult examples to reviewers (Budd et al., 2021; Mosqueira-Rey et al., 2023), while retrieval-augmented and generative systems can supply context or candidate values (Gao et al., 2023; Xu et al., 2024). These capabilities solve complementary inference and interaction problems. They do not, by themselves, constrain the authority of a machine output in the application data model. A high-scoring prediction and a retrieved historical value remain unsafe if either has an unreviewed write path to an authoritative record.

We study the following question: *How can machine perception and optional generative assistance support structured-form digitization while making human authorization an explicit, auditable prerequisite for fact creation?* Auto-Decte addresses this question through two coupled mechanisms. First, *candidate–fact separation* places perception results, retrieved references, and language-model suggestions outside the authoritative fact plane, and makes the boundary executable: each recognition result is persisted as a content-addressed *candidate certificate* bound to its field, evidence hash, producer, and expected fact version, and every confirmation is a per-field attributable human decision over a database-loaded certificate committed in one compare-and-swap-protected transaction. Second, *selective prediction* rejects low-margin recognition results and exposes abstention as an ordinary workflow outcome rather than an exception (Chow, 1970; Geifman and El-Yaniv, 2017, 2019).

The resulting claim is deliberately narrower than "safe AI." Across the application paths exercised in this work, machine modules cannot directly create or revise facts. This architectural property does not guarantee that a human reviewer will detect every error, nor does it establish safety under arbitrary database compromise or malicious operating-system access. It makes each transition into the fact plane explicit, attributable, and traceable to evidence.

This paper makes five contributions:

1. We formalize a four-plane trust model—evidence, candidates, human decisions, and versioned facts—with a machine-to-fact direct-write exclusion invariant scoped to exposed application services, and we define its executable form: content-addressed candidate certificates, per-field attributable decisions, and a compare-and-swap-protected fact transition.
2. We instantiate the model in a runnable Python 3.11 and Streamlit system for fixed industrial forms, combining QR template identity, ArUco localization, selective cell recognition, optional AI review, local retrieval, certificate-bound confirmation, and auditable export.
3. We connect cell recognition to an executable whole-form and application-workflow benchmark and to an authority-integration benchmark that runs the real services over a synthetic fault corpus: 60 evaluation forms, 300 form–condition rows per pipeline variant, 1,200 stored candidates, and 12 pipeline cases with five deterministic corruption and rejection probes.
4. We test the authority boundary through controlled fault families, a candidate-isolation ablation, cross-field and cross-form evidence probes, fail-closed reverse-trace verification over the full certificate, decision, and transition bindings, and a 239-test software verification suite.
5. We report both favorable and unfavorable evidence. HOG–SVM reached 99.81% cell accuracy, whereas complete-form success was 60%

and exposed QR and marker failures under blur and rotation; every injected authority corruption was detected, and no routing advantage is claimed. The architecture, not recognition novelty, is the main contribution.

The remainder of the paper distinguishes related inference and governance work (Section 2), defines the trust invariant (Section 3), describes the implementation and selective rule (Section 4), and presents the deterministic synthetic protocol and results (Sections 5–6). Sections 7–9 delimit industrial relevance and remaining validation needs.

2. Related work

2.1. Document understanding and structured forms

Document layout analysis has progressed from engineered segmentation pipelines to multimodal representation learning (Binmakhashen and Mahmoud, 2019). LayoutLM and its successors jointly model textual, visual, and spatial signals (Xu et al., 2020, 2021; Huang et al., 2022); DocFormer fuses text, vision, and spatial features (Appalaraju et al., 2021); Donut removes a separate OCR stage (Kim et al., 2022); and TrOCR casts text recognition as sequence modeling (Li et al., 2023). PP-OCRv3 represents a highly optimized lightweight OCR pipeline (Li et al., 2022). Recent surveys show that form understanding spans key–value extraction, table structure, handwriting, and heterogeneous layouts (Abdallah et al., 2024). Auto-Decte occupies a narrower regime: fixed, versioned forms whose geometry is known in advance. This restriction permits deterministic localization and makes governance, abstention, and traceability the research focus.

Fiducial markers provide a reproducible geometric reference for this regime. ArUco offers efficient square-marker detection (Romero-Ramirez et al., 2018), and optimized dictionaries increase separation among marker codes (Garrido-Jurado et al., 2016). We use QR identity and four ArUco corners to select a template and normalize perspective; we do not claim a new marker detector.

2.2. Human oversight and selective prediction

Human-in-the-loop machine learning commonly allocates labeling or correction effort where it is most informative or where models are uncertain (Budd et al., 2021; Mosqueira-Rey et al., 2023). Human–AI interaction guidance additionally stresses communicating uncertainty, supporting correction, and preserving user control (Amershi et al., 2019). Auto-Decte adopts these interaction principles but gives review a stronger data-model role: an attributable review event is the only application transition that creates a new fact version.

Reject-option classification dates to the formal trade-off between recognition error and rejection (Chow, 1970). Contemporary selective classifiers jointly optimize prediction and coverage (Geifman and El-Yaniv, 2017, 2019); calibration and conformal methods provide related tools for interpreting or controlling uncertainty (Guo et al., 2017; Angelopoulos and Bates, 2023). We use a simple margin score because it is transparent and reproducible for fixed digit cells. The novelty claim is not the score itself, but its integration with a fact-authority boundary: both accepted and abstained recognition outcomes remain candidates until reviewed.

2.3. Provenance, generative assistance, and authority

Data provenance explains why and where a result arose (Buneman et al., 2001; Herschel et al., 2017); the W3C PROV data model provides interoperable concepts for entities, activities, and agents (World Wide Web Consortium, 2013). Auto-Decte applies the same concern operationally by linking evidence hashes, recognition attempts, review events, record versions, and export batches.

Large language models can generate structured extraction candidates (Xu et al., 2024), and retrieval-augmented generation can ground responses in external material (Gao et al., 2023). Yet grounding does not confer authority: a relevant record may be stale, and a plausible suggestion may still be wrong. We therefore treat retrieval as read-only context and AI output as an append-only suggestion requiring human confirmation. This follows human-centered principles of reliability and control (Shneiderman, 2020) and addresses one point in the broader lifecycle of machine-learning harm (Suresh and Guttag, 2021). The NIST AI Risk Management Framework likewise emphasizes governance and traceability rather than accuracy alone (Tabassi, 2023). Auto-Decte contributes a concrete, testable application invariant for that governance layer.

2.4. Positioning of the contribution

Capability-oriented protection separates possession of authority from a component's descriptive identity or role (Dennis and Van Horn, 1966). Conventional approval, provenance, and access-control mechanisms can record or restrict an action, but they do not by themselves specify how uncertain machine output should enter an operational record. Auto-Decte does not claim a new general access-control model. Its narrower contribution is to couple selective document inference to a durable non-authoritative candidate type, reserve fact creation for a version-checked review capability, and test this separation through executable workflow and fault experiments.

Prior work generally optimizes one of three layers: document inference, uncertainty-aware routing, or human interaction. Auto-Decte connects them through this authority model. Machine output is useful but non-authoritative; review is not merely a correction interface but a state transition; provenance is not merely metadata but the basis of reverse trace. This positioning also explains why a stronger classifier does not invalidate the contribution: any recognizer can replace the current baseline without changing the candidate–fact contract.

3. Trust model and problem formulation

Let a captured form image be evidence $e \in E$, and let M denote the set of machine modules exposed through the application: perception, optional language-model review, and retrieval. A module may derive a candidate $c \in C$ from evidence or existing non-authoritative context. Every persisted candidate carries a *certificate*: a content address computed over its field key, value payload, evidence hash and locator, producer and template identity, selection artifact, target record, expected fact version, and creation time. The certificate is an application-level content-addressed binding artifact used by the authorization workflow, not a PKI certificate, digital identity credential, or authorization token in itself. A human decision $h \in H$ records an actor, a per-field action, an expected record version, a reason, and the referenced certificate. Authoritative state is represented as a sequence of fact versions $F_0, F_1, \ldots$.

The allowed transitions are

$$E \xrightarrow{m} C, \qquad C \xrightarrow{\text{review}} H, \qquad F_{t+1} = g(F_t, H_t), \tag{1}$$

where g validates the expected version and appends, rather than overwrites, the resulting record. The transition is per field: one attributable decision and one fact transition per field, bound to the exact certificate the human relied on, with the certificate's field key, evidence binding, and expected version checked before any write, and the whole version committed by one compare-and-swap on the current fact version. Retrieval can add a reference to C but cannot supply H. The central invariant is

$$\forall m \in M, \quad \neg\big(m \xrightarrow{\text{write}} F\big). \tag{2}$$

In Equation 2, $m \xrightarrow{\text{write}} F$ denotes an exposed application-level direct fact-writing operation. The invariant is an application-architecture property: no exposed machine service possesses such an operation. It is not a claim about protection against privileged database modification, compromised dependencies, or an adversary controlling the host.

In this prototype, human authorization denotes an attributable review action issued through the review capability; reviewer identity is recorded but not cryptographically authenticated. The evaluated invariant concerns capability separation rather than proof of human presence. Figure 1 shows the trust model.

For a recognition input x, a predictor returns a class $\hat{y}(x)$ and a selection score $s(x)$. Given threshold τ, the selection function is

$$a_\tau(x) = \mathbb{1}[s(x) \geq \tau]. \tag{3}$$

For a sample of n labeled cases, coverage and selective risk are

$$\widehat{\text{cov}}(\tau) = \frac{1}{n}\sum_{i=1}^{n} a_\tau(x_i), \tag{4}$$

$$\widehat{R}(\tau) = \frac{\sum_{i=1}^{n} a_\tau(x_i)\mathbb{1}[\hat{y}_i \neq y_i]}{\max(1, \sum_{i=1}^{n} a_\tau(x_i))}. \tag{5}$$

The erroneous acceptance rate uses all cases in the denominator. These empirical measures characterize the recognition gate; they do not alter the authority rule. An accepted recognition is still a candidate, not a fact.

The desired properties are: (P1) direct-write exclusion, Equation 2; (P2) append-only candidate and fact histories whose candidates are content-addressed certificates; (P3) optimistic concurrency, so a stale decision cannot overwrite a newer version; (P4) content-addressed evidence and reverse trace from export to its supporting chain, rebuilt per field as evidence–certificate–decision–transition with any tampered or missing binding reported as incomplete; and (P5) graceful operation when the optional AI adapter is disabled.

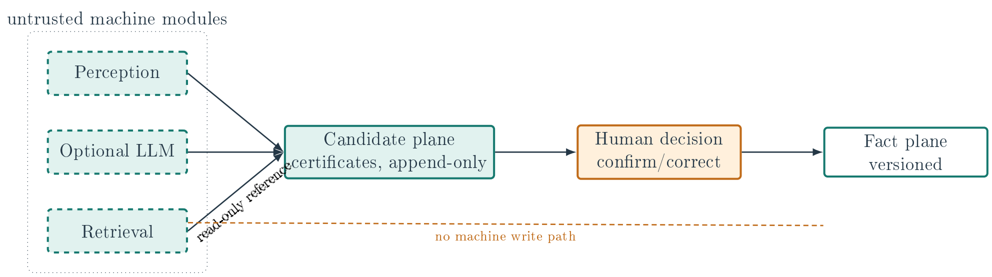

Figure 1: Trust model. Dashed machine components terminate at append-only candidates or read-only references; an attributable human decision is the only exposed path to a versioned fact.

4. System and methodology

4.1. Template-guided perception

Each form template carries a QR payload identifying its template and version and four ArUco corner markers. The system decodes the QR, estimates a homography from detected corners, warps the image to a canonical canvas, and crops fields from versioned template coordinates. This converts a whole-page task into typed cell decisions and preserves the transformation parameters with the recognition attempt.

Digit crops are resized to 48×64 pixels. The built-in comparator binarizes them with Otsu's method and measures normalized mean absolute difference from ten rendered templates. If $d_1(x)$ and $d_2(x)$ are its best and second-best distances, its selection score is

$$s(x) = d_2(x) - d_1(x). \tag{6}$$

The evaluated whole-form configuration instead uses a HOG descriptor with a 48×64 window, 16×16 blocks, 8×8 cells and stride, and nine orientation bins, followed by a linear support-vector machine. Training uses 240 separately seeded rendered cells (eight seeds, three fonts, and ten digits). Its selection score is the difference between the largest and second-largest decision values. Calibration selected $\tau = 0$, so all evaluation predictions were retained; the candidate–fact boundary still requires review before a prediction becomes authoritative. OMR cells use an interior fill ratio: at most 0.15 is unchecked, at least 0.35 is checked, and the interval between them yields selector abstention and is flagged for manual resolution (Figure 2).

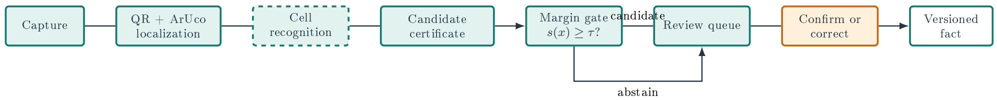

Figure 2: Operational pipeline. Recognition outcomes pass the selector as accepted candidates or abstentions; both enter the mandatory review queue and remain non-authoritative until a confirm/correct action.

4.2. Candidate, decision, and fact services

Evidence is assigned a SHA-256 digest. A recognition attempt atomically persists the evidence row, the attempt, its *candidate certificate*, and an audit event in one transaction; a failure discards the stored crop so nothing referenceable remains. The certificate stores the field key, the canonical value payload, the evidence hash and locator, the producer and template identity, the selection artifact identifier, the target record, the expected fact version, and a content address computed over all of them. The selection artifact identifies the acceptance-threshold policy applied; it is a policy identifier, not a statistical calibration claim. On any subsequent read, the stored content address is re-derived so a modified certificate row is detected.

The optional AI adapter accepts structured context and must return schema-valid review JSON; its prompt contract states that it cannot modify business facts. The default adapter is disabled, returning an explicit `NOT_RUN` status. Local retrieval uses a dependency-free similarity index and exposes results only as references.

The review service confirms one form version under the authority contract. For each field it loads the selected certificate from the database (never from the caller), requires the certificate's field key to equal the mapping key, independently loads the referenced evidence row and requires its owner form and hash to match the certificate, and validates identity, provenance, freshness, version, and content address. Each field yields one attributable decision and one fact transition bound to that certificate; a human-modified machine value derives a manual-entry certificate that inherits the evidence binding and records the machine certificate as lineage parent; an abstained candidate is resolved with an explicit manual-resolution flag. The record version, all decisions, all transitions, derived certificates, and the audit event commit in a single transaction whose first write is a compare-and-swap on the current fact version, so concurrent or stale confirmations fail closed. Version creation enforces a complete per-field transition set over the record's fields;

a transition set that is missing, extra, duplicated, or blank fails closed and is reported per field by the reverse trace. Exports query confirmed records only. Figure 3 shows the reverse-trace chain.

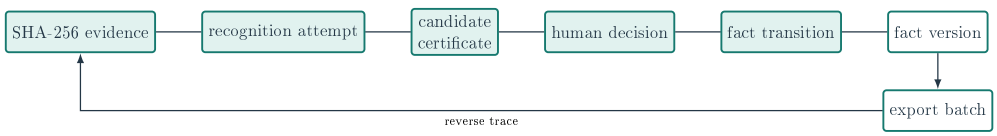

Figure 3: Evidence lineage from content-addressed input to export. Every arrow represents a persisted identifier used by the reverse-trace check; each fact field rebuilds evidence–certificate–decision–transition from the database.

4.3. Implementation

The evaluated artifact is a local single-machine prototype implemented with Python 3.11, Streamlit, SQLite, SQLAlchemy, Alembic, OpenCV, scikit-learn, and an optional provider adapter. Domain, application, adapter, and user-interface layers separate fact-transition services from machine-facing services. The experiment harness invokes these same repositories and application services for import, candidate persistence, confirmation, export, and trace queries.

The system retains local evidence, database, and export files as one backup unit. This local-first operational choice supports recovery in a small-plant setting (Kleppmann et al., 2019), but it is not itself a security boundary. Deployment still requires host access control, encrypted backups, retention rules, and tested restore procedures.

5. Experimental protocol

5.1. Research questions

The evaluation addresses four questions. **RQ1** asks how recognizer choice and risk-targeted selection change coverage and error exposure. **RQ2** asks whether QR identification, marker alignment, digit recognition, and OMR recognition operate together on perturbed whole forms. **RQ3** asks whether injected machine, evidence, and workflow faults can change an existing fact or corrupt its reverse trace through exposed application paths, including cross-field certificate substitution, cross-form evidence, tampered bindings, missing transitions, and record-version rebinding. **RQ4** asks whether import–candidate–review–export operations produce versioned facts and complete, fail-closed reverse traces without the AI adapter, under an executable benchmark whose outputs are byte-identical across repeated runs.

5.2. Synthetic recognition benchmark

Production forms can contain employee identifiers, payroll-adjacent values, and commercially sensitive production data, and no production-form corpus was available to this study under an authorized research-use protocol. Production records were therefore neither used in the public benchmark nor sent to external AI services. Instead, the package deterministically renders synthetic form cells with known labels. This decision makes the benchmark distributable and repeatable but limits ecological validity.

The benchmark contains 4,350 image rows. Seeds 101 and 202 define 1,740 calibration rows; disjoint seeds 303, 404, and 505 define 2,610 evaluation rows. The evaluation split contains 90 seed–font–digit base units, each observed under 29 clean or perturbed conditions. Consequently, rows are disjoint from calibration by seed but are not mutually independent within the evaluation split. Training data for the learned baseline use separate seeds. Perturbations comprise clean images plus Gaussian blur, Gaussian noise, brightness reduction, rotation, perspective distortion, JPEG compression, and occlusion at predefined severities. The implementation records every raw prediction, score, label, model, perturbation, severity, and seed.

Thresholds from 0 to 0.10 are swept in increments of 0.005. For each selective model, the operating threshold is selected *only on calibration rows* as the lowest grid value satisfying a target empirical selective risk of at most 1%, thereby retaining the highest calibration coverage among qualifying thresholds. The selected value is then frozen for evaluation. We report coverage, accepted accuracy, selective risk, erroneous acceptance rate, and abstention rate. Selector abstention is distinct from mandatory authorization review: both accepted and abstained machine outputs remain non-authoritative candidates until an attributable review action occurs. Uncertainty is estimated with 2,000 cluster-bootstrap replicates that resample the 90 seed–font–digit base units and retain all 29 rows within each sampled unit. A row-level Wilson interval is retained only as a descriptive comparison because it assumes independent Bernoulli trials.

Three available recognition paths are compared: the Auto-Decte template matcher, the same template decision forced to predict whenever possible, and a HOG feature extractor with a linear support-vector machine. A Tesseract path is capability-gated. The executable was unavailable in the recorded environment, so no Tesseract score is reported.

5.3. Whole-form benchmark and component-suppression analysis

The form generator produces a 700×1000-pixel page with a template-and-version QR payload, four ArUco corners, ten digit fields, and ten OMR fields. Two templates differ in identifier, version, and font. Ten calibration seeds and 30 disjoint evaluation seeds yield 20 and 60 base forms, respectively. Each base form is evaluated under clean, Gaussian-blur, rotation, perspective, and JPEG conditions. The evaluation therefore contains 300 form–condition rows per pipeline variant.

The primary endpoint is complete-form success: correct template identity, successful alignment, ten accepted and correct digits, and ten accepted and correct OMR values. Secondary endpoints report each component separately. We compare the full pipeline with variants that suppress QR identity or ArUco alignment. Under this endpoint, removing either required component necessarily prevents complete-form success; the field-level metrics show whether component removal also propagates to recognition. Separately, each clean evaluation form traverses image import, storage of 20 recognition candidates, a programmatically invoked review-service confirmation against known synthetic labels, creation of fact version 1, XLSX export, and reverse trace. This is an executable workflow test, not a study of human reviewer behavior.

5.4. Authority-integration benchmark

A second executable benchmark runs the integrated services (import, recognition with certificate creation, certificate-bound confirmation, reverse trace) over a synthetic fault corpus with a scripted reviewer and a fixed injected clock and seed, so repeated runs are byte-identical. Twelve pipeline cases exercise the normal path; five deterministic probes exercise the authority boundary: (i) a certificate bound to one field supplied under another field must be rejected before any write; (ii) evidence owned by another form must be rejected; (iii) a transition's producer binding is tampered after confirmation; (iv) one field's transition is deleted from a two-field version; and (v) a transition's record-version row identifier is rebound to a different, existing version row, which the database foreign key alone cannot distinguish. Metrics are counted from the database: corruption detection rate (probes iii–v), invalid-transition rejection rate (probes i–ii), valid-certificate coverage (the twelve normal cases), reverse-trace completeness over the valid traced forms, with corruption detection and invalid-transition rejection reported separately, stale-transition containment, and scripted-reviewer wrong-value retention (the share of injected wrong values the scripted reviewer accepts; recorded in the manifest under the key `silent_fault_escape_rate`). The trace contract is also exercised by the test suite with foreign keys disabled, altering every duplicated binding field independently and asserting the specific failure.

5.5. Fault injection and resilience

Six controlled cases probe distinct boundaries: a wrong recognition candidate, a wrong language-model suggestion, an irrelevant retrieval result, a stale human-review replay, duplicate evidence, and attempted direct fact writes. The last case tests four fact-like method names against each of the recognition, AI-review, and retrieval services (12 paths). A case passes only when the existing fact remains unchanged and an audit outcome is recorded; stale review and duplicate evidence must additionally raise their designated exceptions, and direct-write methods must be absent.

A candidate-isolation ablation compares the normal wiring with a test-only unsafe path that grants the injected machine value access to the review service. A randomized stress experiment then runs 100 seeded trials for each of five fault families (500 total). Resilience checks exercise import, programmatic review confirmation, export, reverse trace, duplicate rejection, stale-version rejection, and operation with AI disabled. Service latency is measured over 30 repetitions after five warm-ups; because it excludes perception, browser rendering, storage-device variance, and human review, it is reported only in the supplementary material.

5.6. Reproducibility controls

The experiment code fixes seeds and records dependency versions, emits JSON before figures or tables, and writes a manifest containing SHA-256 hashes, commands, environment metadata, source commit, and working-tree status. The authority-integration benchmark additionally fixes the evaluation clock and writes an immutable manifest (source commit, standard work-tree cleanliness, config hash, seeds, package versions, artifact hashes) that excludes itself; two independent runs produce byte-identical payloads and manifests, and the artifact hashes are independently recomputed. Paper tables are generated from the same JSON read by the plots. The final code quality gate comprises the automated test suite (239 tests in the verified integrated state, pinned to the source commit reported in the supplementary material), Ruff static checks, and mypy type checks; the test count is software verification information and is not presented as a scientific sample size. The evaluation is synthetic by design; a production claim would require a separate, access-controlled protocol with double transcription and adjudication.

DeepSeek Harness and OpenAI Codex were used as AI-assisted development tools for code drafting, refactoring, and test suggestions. The authors selected the experimental design, reviewed the resulting changes, executed the complete quality gate, and verified the reported artifacts and references. No production record was supplied to these tools. The corresponding manuscript-preparation use is disclosed separately in the declaration section.

6. Results

6.1. RQ1: recognition and selective risk

The template matcher required a conservative operating point. Calibration selected $\tau = 0.08$; on the 2,610 evaluation rows, it accepted 292 predictions and abstained on 2,318. Coverage was 11.19% (cluster-bootstrap 95% interval 5.86%–16.97%), with no accepted error observed. The corresponding cluster-bootstrap interval for accepted accuracy was 100%–100%; this degenerate interval reflects the deterministic sample and does not establish a zero population error rate.

The HOG–linear-SVM changed the operating trade-off. Its calibration split already met the 1% risk target at $\tau = 0$, so the frozen selector accepted all 2,610 evaluation rows. It correctly classified 2,605, giving 99.81% accepted accuracy and 0.19% selective risk. Cluster resampling placed the risk interval at 0.038%–0.346% and the accepted-accuracy interval at 99.62%–99.96% (Table 1). The result supports HOG–SVM as the evaluated digit recognizer; it does not support a claim of recognition novelty. Figure 4 compares the two calibration-selected operating points.

The template gate remains informative as a stress case. Figure 5 shows that its full-coverage end incurred roughly 25% selective risk; the first threshold with zero observed accepted error was $\tau = 0.07$, at 14.83% coverage. Severe perspective distortion and occlusion reduced its coverage to 3.33%. Thus, selection prevented many weak template predictions from being accepted by the selector, but the stronger HOG representation removed most of that synthetic-regime trade-off.

Table 1: Calibration-selected operating points on the seed-disjoint evaluation split. Confidence intervals reported in the text resample 90 base units rather than 2,610 rows.

Model	Cov. (%)	Acc. (%)	Risk (%)	Abstain (%)
Template matcher	11.2	100.00	0.00	88.8
HOG–SVM	100.0	99.81	0.19	0.0

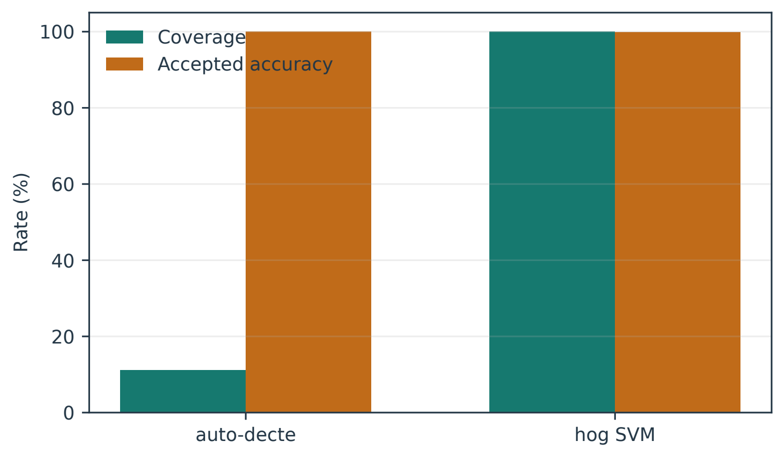

Figure 4: Coverage and conditional accepted accuracy for the two calibration-selected recognizer–gate combinations.

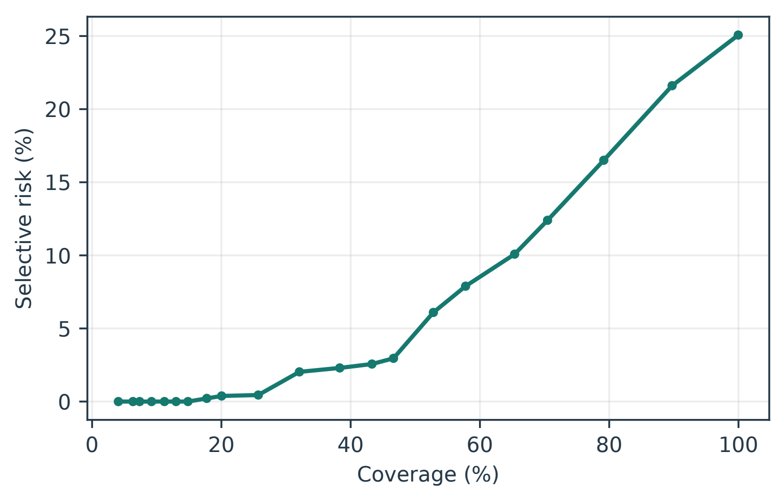

Figure 5: Template-matcher coverage–risk curve. The dashed operating point was selected on calibration data before evaluation.

6.2. RQ2: whole-form integration

Across the 300 evaluation form–condition rows, template identification succeeded on 80% and ArUco alignment succeeded on 80%; both succeeded together on 180 rows (60%). Across all conditions, digit accuracy was 86.03%, OMR accuracy was 100%, and complete-form success was 60% (Table 2). All 60 clean, 60 perspective, and 60 JPEG rows were completely correct. Gaussian blur preserved alignment and 100% field accuracy but prevented QR identification on all 60 rows. Rotation preserved QR identification but prevented marker alignment on all 60 rows and reduced digit accuracy to 30.17%. These complementary failures explain why the two marginal component rates exceed their joint rate and why the complete endpoint is 60%.

Table 2: Whole-form evaluation over 300 form–condition rows per variant. Complete success requires template identity, alignment, and all 20 fields correct.

Variant	QR (%)	Align (%)	Digit (%)	OMR (%)	Complete (%)
full_pipeline	80.0	80.0	86.0	100.0	60.0
without_qr	0.0	80.0	86.0	100.0	0.0
without_aruco	80.0	0.0	69.4	99.0	0.0

Suppressing QR identification left field recognition unchanged but made the complete endpoint unattainable. Suppressing ArUco alignment reduced digit accuracy from 86.03% to 69.43% and OMR accuracy from 100% to 99.03%; complete-form success again fell to zero. Figure 6 separates the component rates so the endpoint's conjunctive definition is explicit.

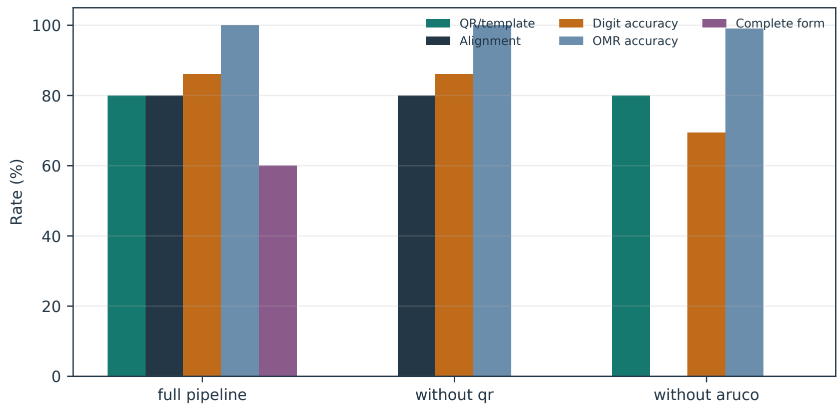

Figure 6: Whole-form component-suppression analysis. Removing a component required by the endpoint forces complete success to zero; field metrics show the additional recognition effect of missing alignment.

6.3. RQ3: authority-boundary faults

All six controlled cases left the pre-existing fact unchanged and produced complete audit evidence. Wrong recognition and language-model outputs remained candidates, irrelevant retrieval remained a reference, stale review and duplicate evidence raised their designated exceptions, and all 12 attempted direct fact-write paths were absent from the three machine-facing services.

The candidate-isolation ablation supplied an incorrect value of 999. Under normal wiring, recording the candidate did not change the fact. Under test-only unsafe wiring that granted the same value access to the review service, the fact changed. This contrast locates containment in capability separation rather than in the injected value or recognizer behavior.

Randomized stress testing repeated five fault families 100 times each. All 500 trials were contained (Table 3). This expands the evidence beyond one example per family, although it remains an application-path test rather than a penetration test, formal proof, or assessment of host compromise.

Table 3: Seeded randomized trust-boundary stress test.

Fault family	Contained	Trials
wrong_recognition	100	100
wrong_llm_suggestion	100	100
irrelevant_retrieval	100	100
stale_human_replay	100	100
duplicate_evidence	100	100

The authority-integration benchmark runs the integrated services with certificates, per-field confirmation, and the database-rebuilt reverse trace. Over 12 pipeline cases with a scripted reviewer, all 12 valid certificate confirmations succeeded (valid-certificate coverage 1.0) and produced complete reverse traces (valid trace completeness 12/12). Every injected authority corruption was detected: a tampered transition producer, a deleted field transition, and a transition rebound to another existing version row each produced an incomplete trace (corruption detection 3/3); the aggregate reverse-trace completeness over all 15 traced forms was therefore 12/15, counting the three intentionally corrupted traces as incomplete. Cross-field certificate substitution and cross-form evidence were both rejected before any write (invalid-transition rejection 2/2), and a stale replay was contained. The scripted reviewer accepted three of four injected wrong machine values (scripted-reviewer wrong-value retention 0.75); this is scripted behavior on synthetic faults and is not reported as human performance. The test suite extends the same checks with foreign keys disabled, altering each duplicated binding independently and asserting the specific failure, including the record-version row binding that a foreign key alone cannot distinguish.

6.4. RQ4: executable workflow and reproducibility

Each of the 60 clean evaluation forms traversed the complete application workflow. The runner imported the original image, stored 20 recognition candidates, programmatically submitted known synthetic labels through the review service, created fact version 1, exported an XLSX file, and queried the reverse trace. All 60 workflows succeeded, accounting for 1,200 persisted candidate attempts. This verifies the software path but not human-review effectiveness. AI-disabled operation, duplicate rejection, stale-version rejection, and reverse tracing also passed.

The integrated verification run passes 239 automated tests together with Ruff and mypy checks; the count is software verification information and not a scientific sample size. The authority-integration benchmark was run twice to separate scratch directories: the payload and manifest are byte-identical, the manifest records the final source commit with a clean worktree and an independently recomputed artifact hash. Service microbenchmarks and the full perturbation table are supplied in the supplementary material because they do not measure end-to-end reviewer latency.

7. Industrial context and intended use

The system was motivated by paper-based production records in a bamboo-processing setting, where form values can be adjacent to employee identity, output accounting, and payroll decisions. The intended user is a reviewer who receives a digitized form, inspects machine candidates alongside image evidence, confirms or corrects values, and exports only confirmed records. The prototype demonstrates that this workflow can be represented in software, but the paper does not claim a measured deployment effect on cost, time, accuracy, wages, or organizational outcomes.

No production-form corpus was available to this study under an authorized research-use protocol. We therefore used deterministic synthetic forms for controlled and reproducible mechanism evaluation. Privacy, commercially sensitive content, and authorization requirements further constrain the acquisition and public release of such data. Synthetic forms preserve known labels, controlled perturbations, and distributable evidence. They do not reproduce handwriting variation, capture habits, paper wear, device diversity, or reviewer behavior in the plant.

For production adoption, the authority model should be embedded in a broader control system: least-privilege host access, encrypted storage and backups, retention schedules, separation of reviewer and payroll roles where required, audit review, incident response, and periodic sampling of confirmed records. Candidate–fact separation reduces one route by which machine error becomes operational truth; it does not replace these organizational controls.

8. Discussion

The experiments distinguish predictive performance from governance performance. HOG–SVM reduced cell error to 0.19% in the synthetic evaluation, but five mistakes remained. The authority boundary addresses the destination of such errors: even a selected prediction is persisted as a candidate until an attributable, version-checked review creates a fact. Stronger perception therefore reduces reviewer workload without making candidate–fact separation redundant.

Recognizer choice also changed the role of selective prediction. The template matcher needed to abstain on 88.81% of evaluation rows to achieve zero observed accepted errors, whereas HOG–SVM satisfied the calibration risk target at full coverage. This difference argues against treating a threshold as a fixed safety setting. Selection must be recalibrated after changes in forms, printers, cameras, or recognizers, and a deployment threshold should reflect the relative cost of an unreviewed error, reviewer time, and delayed processing.

The whole-form benchmark exposed failures that the cell benchmark could not. Digit and OMR recognition were perfect under blurred forms once fields were localized, yet QR identification failed, so no complete record could be produced. Under rotation, ArUco alignment failed and digit accuracy fell sharply. These observations make the engineering priorities concrete: QR preprocessing or fallback classification is needed for blur, and marker detection or orientation normalization is needed for rotation. The 60% complete-form result is consequently more informative than the 99.81% isolated-cell score for the current pipeline.

The controlled fault cases, candidate-isolation ablation, and 500 randomized trials test the causal role of capability separation. The incorrect value was harmless when stored through a candidate-only interface and changed the fact when the experiment deliberately exposed the review capability. This is stronger evidence than an architecture diagram alone, but weaker than formal verification of the direct-write exclusion property, database privilege separation, or adversarial assessment of every UI and migration path.

The authority-integration benchmark tests the boundary in its executable form. Certificates bind a candidate to its field, evidence, and expected version, so a certificate moved to another field or confirmed against another form's evidence is rejected before any write. Confirmations commit per-field decisions, transitions, and the record version in one compare-and-swap transaction, so concurrent and stale confirmations fail closed. Reverse traces are rebuilt from the database per field rather than assumed, and the benchmark shows that tampered producers, deleted transitions, and record-version rebinding—including rebinding to another existing row that the database foreign key still accepts—are reported as incomplete. The measured wrong-value retention of 0.75 belongs to the scripted reviewer and is deliberately not interpreted as human error rates; that question is deferred to the controlled human-review study. The authority contract prevents machine-only fact creation; it does not convert human authorization into a correctness guarantee.

The architecture remains model-agnostic. A conformal or learned selector can replace the decision-margin rule, and different retrieval or language models can supply assistance, provided that each terminates outside the fact-writing interface. This separation allows model procurement to change while preserving a stable audit question: which service can invoke the fact-transition operation?

The next empirical study should use a blinded, double-entered corpus of real forms with adjudicated ground truth and subgroup reporting by template, device, writing style, and degradation. A second study should randomize review interfaces—recognition only, retrieval-assisted, and AI-assisted—and measure review time, correction accuracy, automation bias, and inappropriate suggestion acceptance. Those studies would test decision quality and operational value, which synthetic execution cannot establish.

9. Limitations and threats to validity

External validity. All quantitative recognition results use deterministic synthetic cells. They do not represent handwriting, Chinese text, signatures, paper damage, camera diversity, or the joint distribution of production forms. The privacy rationale for withholding production data is legitimate, but it does not turn synthetic evidence into field evidence.

Statistical validity. Threshold selection uses calibration seeds disjoint from evaluation seeds, reducing direct evaluation leakage. Nevertheless, calibration and evaluation arise from the same renderer and perturbation generator. The 2,610 cell rows are 29 conditions applied to only 90 base units; cluster bootstrap respects this grouping but cannot create diversity absent from the renderer. The 60 whole-form base units likewise share two templates and one generator. Confidence intervals therefore quantify resampling uncertainty within the synthetic design, not domain shift to production forms.

Construct validity. The trust tests operationalize containment as an unchanged fact plus complete audit evidence. This captures the stated application invariant but not human susceptibility to misleading suggestions, correctness of a later human decision, confidentiality, availability, or resistance to host compromise. The 12 direct-write probes are negative interface checks for four fact-like method names on three machine-facing services; they do not exhaust indirect repository references, migrations, administrative database access, or future UI wiring. The authority-integration evidence covers the tested corruption families—cross-field and cross-form substitution, tampered bindings, missing transitions, and record-version rebinding—and does not generalize to arbitrary attacks. The scripted reviewer in the authority benchmark is not a human reviewer, and its 0.75 wrong-value retention is not a human error estimate. "Trust-constrained" should not be read as a general security certification.

Baseline validity. HOG–SVM is strong on this renderer and may benefit from the constrained visual regime. Tesseract was not installed and was excluded rather than simulated. Broader OCR and document-model comparisons on real data remain necessary.

Implementation scope. The evaluated artifact is a local single-machine prototype with a Streamlit interface. The present experiments do not establish concurrent multi-user behavior, access-control enforcement, encrypted deployment, or resistance to a compromised host. Service timings are microbenchmarks, not end-to-end latency.

Reproducibility. Seeds, code, raw predictions, summary JSON, plots, tables, and hashes are supplied. The authority-integration benchmark fixes its clock and seed, and two independent runs produced byte-identical payloads and manifests with independently recomputed hashes. Reproduction still depends on compatible OpenCV and system behavior. The automated test count (239 in the verified state) is software verification information, not a scientific sample size. The public package intentionally excludes sensitive plant records.

10. Conclusion

Auto-Decte treats machine outputs as evidence-linked candidates rather than authoritative facts. Within the exposed application services, perception, retrieval, and optional language-model review have no direct fact-writing path; a version-checked, attributable human decision over a content-addressed certificate is required. HOG–SVM achieved 99.81% cell accuracy, but the full pipeline completed only 60% of perturbed whole forms because blur defeated QR identity and rotation defeated marker alignment.

In the integrated system, confirmations commit per-field decisions, transitions, and the record version in one compare-and-swap transaction, and reverse traces rebuilt from the database reported every injected corruption—tampered producers, deleted transitions, and record-version rebinding—as incomplete. All 60 clean evidence-to-export workflows were traceable, 12 direct-write attempts were unavailable, and 500 randomized faults were contained without changing existing facts.

The evidence supports the feasibility of the architecture and its executable invariants, not production effectiveness or universal safety. Real-form validation and controlled human-review studies are the necessary next steps. The central design lesson is portable: improve models when possible, but design the data path so that model confidence is never confused with authority.

Declarations

Data and code availability. The reproducible synthetic benchmark, raw predictions, summary artifacts, figure-generation code, application source, and frozen evaluation snapshot are provided in the accompanying reviewer artifact. No production-form corpus was used or included in this study; such records may contain employee identity, payroll-adjacent values, and commercially sensitive production information.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work, the authors used DeepSeek Harness and OpenAI Codex to assist with drafting, content organization, language editing, and AI-assisted code development. The authors reviewed, edited, and verified the resulting text, code, analyses, references, and claims, and take full responsibility for the content of the publication. No generative-AI-created or generative-AI-altered image is included in this submission; diagrams are author-controlled vector graphics and quantitative plots are generated programmatically from the reported experiment artifacts.

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